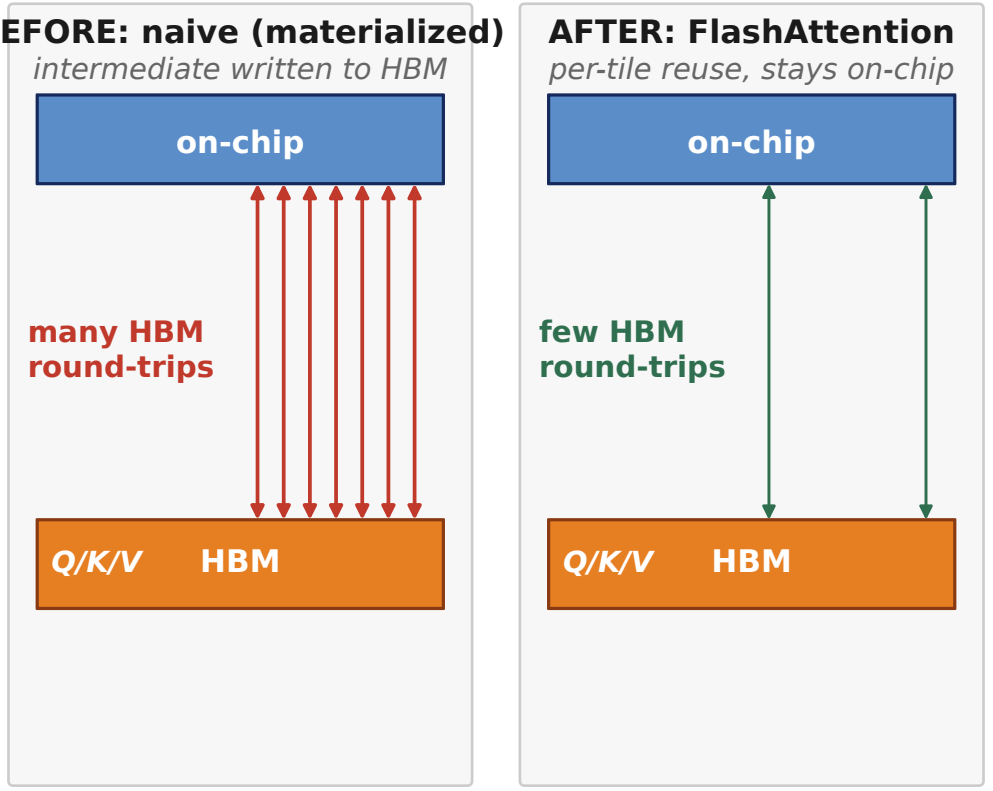
FlashAttention: the same math, far less HBM traffic



O(L²) HBM traffic: L×L intermediate written + every query re-reads the K/V rows

K/V reused per tile — no single global K/V pass; no L×L intermediate at HBM

same $Q \cdot K^T \cdot V$ math, same result (up to last-bit rounding) — the gain is fewer HBM round-trips, not cheaper attention

K/V on-chip reuse is per-tile (per Q-block), not a single global K/V pass.
I/O bound: FlashAttention — Tri Dao et al., arXiv:2205.14135.